

User Manual

AI-Based HR Resume Recommender System

Version 1.0

Final Year Project — Computing Project

System	AI-Based HR Resume Recommender System
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1. Introduction

1.1 Purpose of This Document

This User Manual provides step-by-step guidance for installing, configuring, and using the AI-Based HR Resume Recommender System. It is intended for three audiences: HR professionals who will use the HR Portal to manage jobs and review candidates; job candidates who will use the Candidate Portal to apply for positions; and system administrators or developers who need to set up and run the application.

1.2 System Overview

The AI-Based HR Resume Recommender System is a full-stack web application that automates the resume screening and candidate ranking process for Human Resource (HR) professionals. It eliminates the need for manual resume reading by using Natural Language Processing (NLP) to extract skills and education from uploaded resumes and produce a ranked, scored candidate list for every job posting.

The system consists of three independently running services:

- Frontend (port 5500) — HR Portal and Candidate Portal, built with HTML5, CSS3, and JavaScript
- Backend (port 5000) — RESTful API built with Node.js and Express.js, connected to MongoDB
- AI Module (port 8000) — Python/FastAPI microservice for resume parsing, skill extraction, and scoring using spaCy and TF-IDF

1.3 Key Features

HR Portal:

- Register and log in as an HR user
- Create, edit, and delete job postings with required skills, education level, and match score threshold
- View AI-ranked candidates per job with overall match score, skill breakdown, and education badge
- Update candidate statuses: Shortlisted, Reviewed, Rejected
- Export shortlisted candidates to CSV
- Dashboard with real-time statistics, donut chart, and status bars

Candidate Portal:

- Register and log in as a candidate
- Build a profile with education level and skills
- Upload a resume (PDF, DOCX, or TXT)
- Browse active job listings and apply
- Track application status and view AI match score for each application

1.4 Intended Users

HR Personnel	Uses the HR Portal to post jobs, view AI-ranked candidates, and manage shortlisting decisions
Job Candidates	Uses the Candidate Portal to register, upload their resume, apply for jobs, and track application outcomes
Administrator	Responsible for installing and running all three services; requires basic command-line knowledge

2. System Requirements & Installation

2.1 Prerequisites

Before installing the system, ensure the following software is installed on your machine:

Node.js	v18 or higher — https://nodejs.org
npm	Included with Node.js
Python	3.10 or higher — https://www.python.org
MongoDB	Local instance (Community Edition) — https://www.mongodb.com/try/download/community
Git	For cloning the repository — https://git-scm.com
Web Browser	Google Chrome or Mozilla Firefox (latest version recommended)

 **Tip:** MongoDB Compass (GUI) is recommended for inspecting the database during development but is not required to run the system.

2.2 Cloning the Repository

Open a terminal and run the following command to clone the project:

```
git clone https://github.com/ImanHaqimi/hr-resume-recommender.git
cd hr-resume-recommender
```

The project contains two top-level directories:

- /backend — Node.js/Express server that also serves the frontend
- /ai-module — Python/FastAPI AI microservice

 **Note:** There is no separate frontend server. The backend serves all frontend HTML, CSS, and JavaScript files directly from port 5000.

2.3 Step 1 — Start MongoDB

Ensure MongoDB is installed and running on your machine before starting the other services.

```
# Windows (if MongoDB is installed as a service)
net start MongoDB

# Or start manually
mongod
```

 **Tip:** On macOS/Linux, MongoDB can be started with: `brew services start mongodb-community` or `sudo systemctl start mongod`

2.4 Step 2 — Start the Backend

The backend server also serves the frontend — no separate frontend server is needed.

Step 1 Navigate to the backend directory:

```
cd backend
```

Step 2 Create a .env file in the /backend directory with the following content:

```
MONGO_URI=mongodb://localhost:27017/hr_recommender
JWT_SECRET=your_jwt_secret_key_here
AI_MODULE_URL=http://localhost:8000
PORT=5000
```

⚠ Important: Replace your `_jwt_secret_key_here` with a strong, unique secret string. Never commit your .env file to version control.

Step 3 Install Node.js dependencies (first time only):

```
npm install
```

Step 4 Start the development server (with auto-reload):

```
npm run dev
```

✅ **Backend + Frontend running at: <http://localhost:5000>**

2.5 Step 3 — Start the AI Module

Open a new terminal window for the AI module.

Step 1 Navigate to the AI module directory:

```
cd ai-module
```

Step 2 Install Python dependencies (first time only):

```
pip install -r requirements.txt
```

Step 3 Start the AI service with auto-reload:

```
python -m uvicorn main:app --reload --port 8000
```

✅ **AI Module running at: <http://localhost:8000>**

📖 **Interactive API docs: <http://localhost:8000/docs>**

2.6 Service Summary

Once both services are running, the system is fully operational. You only need two terminal windows:

Frontend + Backend	http://localhost:5000
AI Module	http://localhost:8000
AI API Docs	http://localhost:8000/docs
MongoDB	mongodb://localhost:27017/hr_recommender

💡 **Tip:** Keep two terminal windows open simultaneously — one for the backend (`npm run dev`) and one for the AI module (`uvicorn`). MongoDB must also be running in the background.

3. Getting Started

3.1 Accessing the System

Once all three services are running, open your web browser and navigate to:

`http://localhost:5000`

You will be directed to the login page. From here, you can log in as an existing user or register a new account.

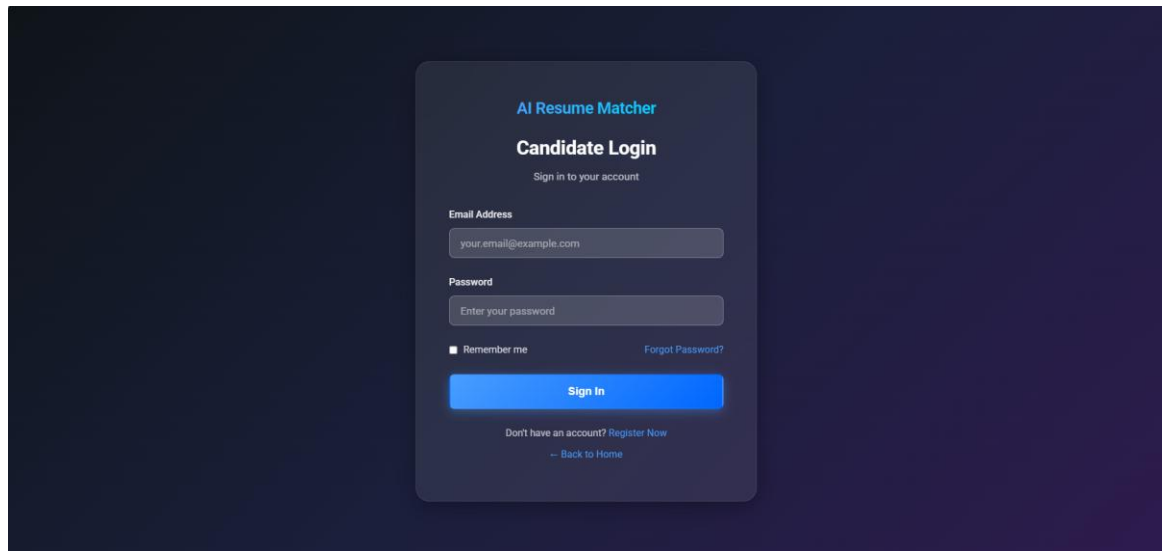


Figure: Login page — `http://localhost:5000`

3.2 Registering an Account

Click the Register link on the login page. Complete the registration form:

Full Name	Your full name
Email	A valid email address (used to log in)
Password	A password of your choice (minimum 6 characters recommended)
Phone	Your contact number

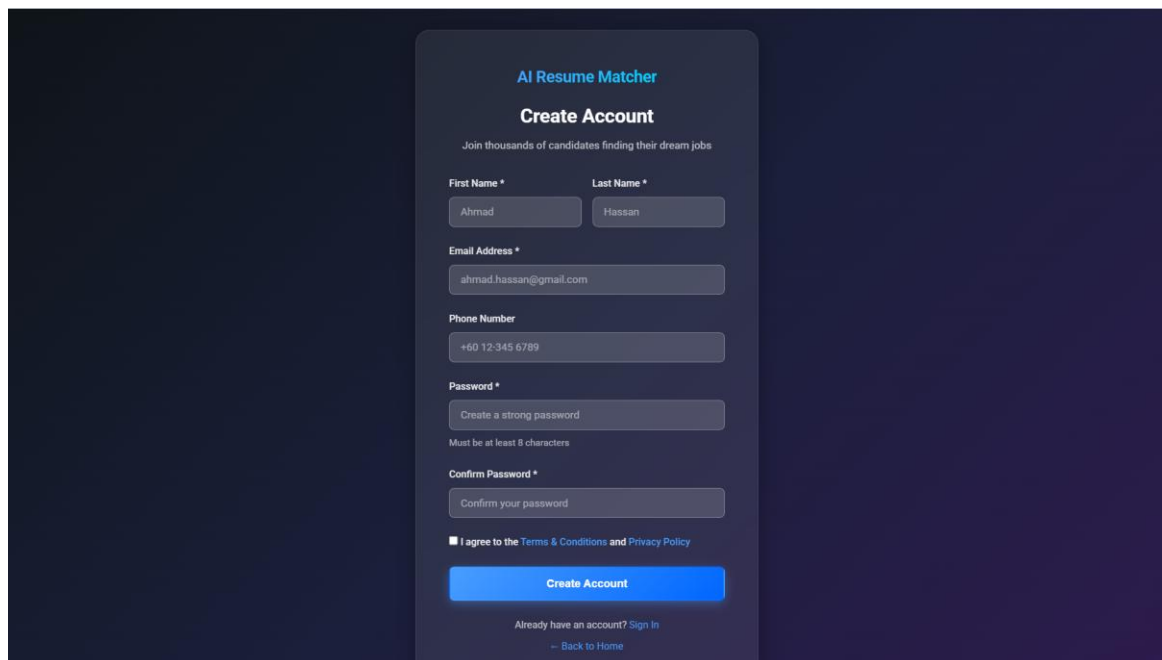
The image shows a registration form for 'AI Resume Matcher'. The form is titled 'Create Account' and includes a sub-header 'Join thousands of candidates finding their dream jobs'. The form fields are: First Name (Ahmad), Last Name (Hassan), Email Address (ahmad.hassan@gmail.com), Phone Number (+60 12-345 6789), Password (Create a strong password), and Confirm Password (Confirm your password). There is a checkbox for 'I agree to the Terms & Conditions and Privacy Policy' and a 'Create Account' button. At the bottom, there are links for 'Already have an account? Sign In' and 'Back to Home'.

Figure: Registration form

3.3 Logging In

Step 1 Enter your registered email address.

Step 2 Enter your password.

Step 3 Click Log In.

Upon successful login, you will be redirected to your role-specific portal — the HR Dashboard for HR users, or the Job Listings page for candidates.

4. HR Portal

The HR Portal is accessible only to users registered with the HR role. It provides tools for managing job postings, reviewing AI-ranked candidates, and exporting shortlists.

4.1 HR Dashboard

After logging in, HR users land on the Dashboard — the central overview page.

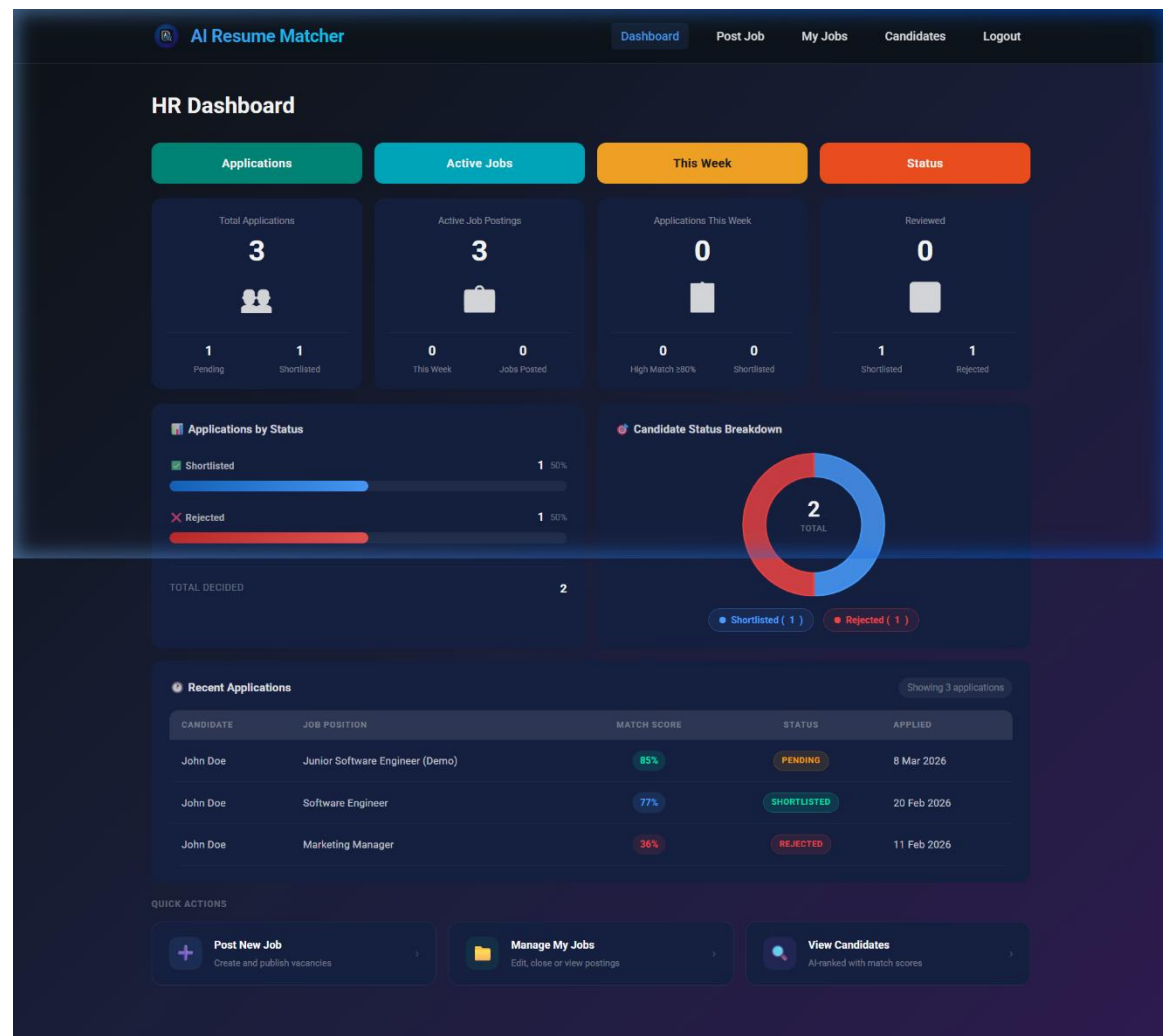


Figure: HR Dashboard — summary cards, donut chart, and status bars

The Dashboard displays:

- Summary Cards — Total Jobs Posted, Total Applications Received, Total Shortlisted, and Total Rejected, updated in real time
- Donut Chart — Visual breakdown of Shortlisted vs Rejected outcomes across all jobs
- Status Bars — Per-job progress bars showing the screening status distribution at a glance
- Recent Activity — A log of the most recent status changes and applications received

4.2 Managing Job Postings

4.2.1 Viewing All Jobs

Click Manage Jobs in the sidebar. All job postings created by your account are listed here, each showing the job title, number of applicants, required skills (as tags), and action buttons.

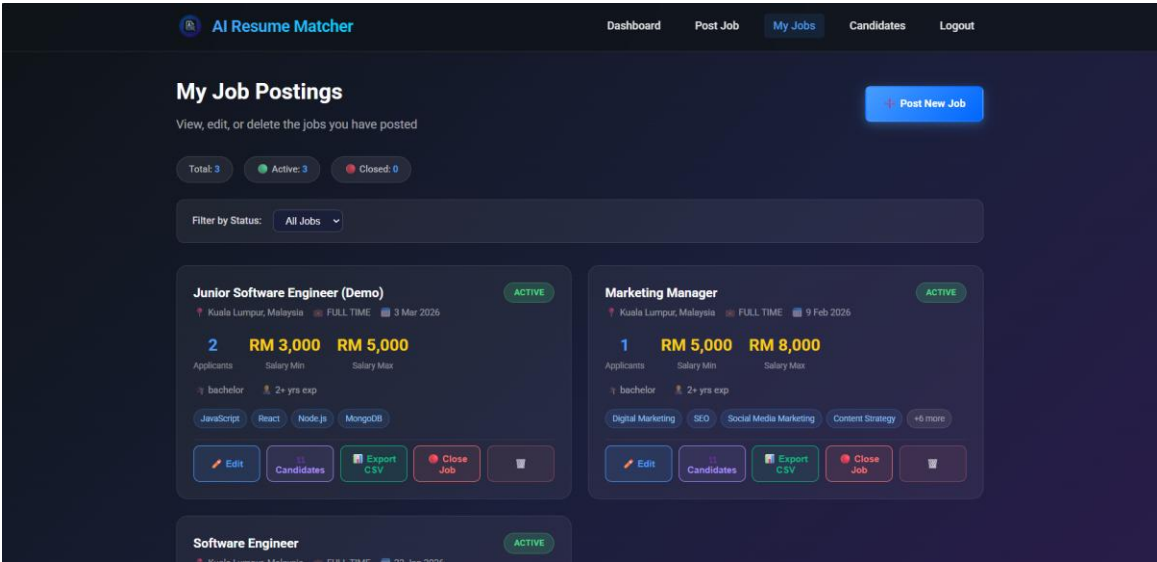


Figure: Manage Jobs page — job list with skill tags and action buttons

4.2.2 Creating a New Job

Step 1 Click the + New Job button on the Manage Jobs page.

Step 2 Fill in the job details:

Job Title	The name of the position (e.g. Junior Software Engineer)
Job Description	A detailed description of the role and responsibilities
Required Skills	Enter each skill and press Enter or comma to add it as a tag (e.g. Python, JavaScript, MongoDB)
Required Education	Select the minimum education level: Diploma, Bachelor's, Master's, or PhD
Min. Match Score (%)	Candidates scoring below this threshold will be automatically filtered out. Set to 0 to show all candidates.

Step 3 Click Save Job. The job will appear in your Manage Jobs list and be visible to candidates immediately.

Figure: Create Job form — with required skills and education threshold fields

4.2.3 Editing a Job

Step 1 On the Manage Jobs page, click the Edit (pencil) icon next to the job you want to update.

Step 2 Modify the relevant fields.

Step 3 Click Save Changes.

4.2.4 Deleting a Job

Step 1 On the Manage Jobs page, click the Delete (bin) icon next to the job.

Step 2 Confirm the deletion in the prompt.

⚠ Warning: Deleting a job permanently removes it and all associated applications. This action cannot be undone.

4.3 Viewing AI-Ranked Candidates

This is the core feature of the HR Portal. For any job posting, the system automatically ranks all applicants using the AI scoring engine.

Step 1 From Manage Jobs, click the View Candidates button for a job, or navigate to Candidates in the sidebar and select the job.

Step 2 The candidates page loads, displaying all applicants ranked from highest to lowest match score.

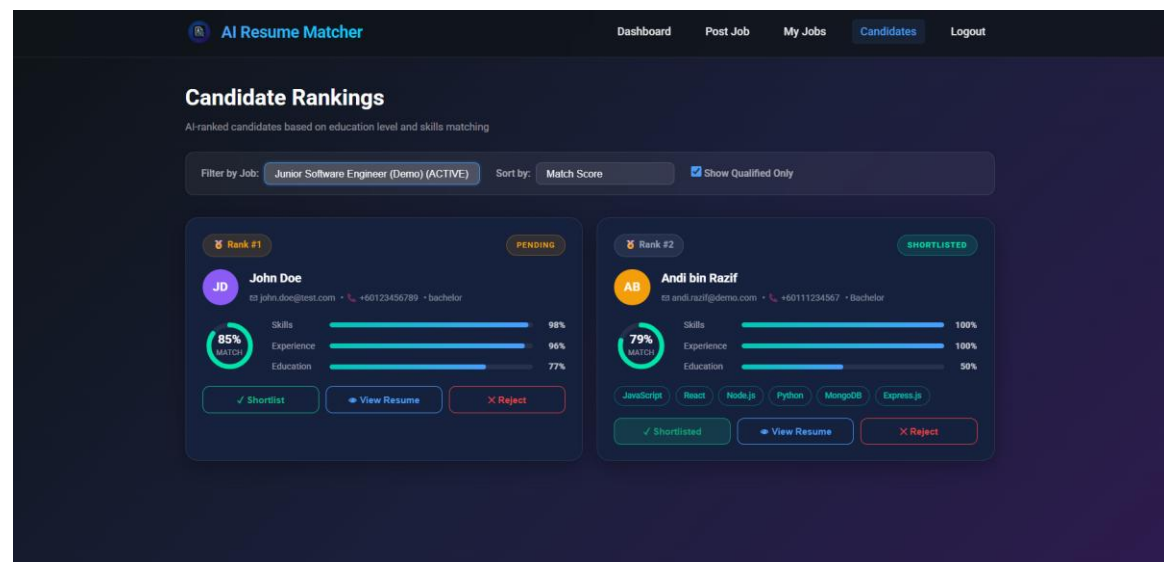


Figure: Candidate Ranking page — AI-ranked list with match scores, skill breakdown, and status controls

Each row in the ranking table shows:

- Rank — Position in the AI-ranked list (1 = best match)
- Candidate Name and Email
- Overall Match Score (%) — Colour coded: Green ($\geq 70\%$), Amber (40–69%), Red ($< 40\%$)
- Education Level — Badge indicating the candidate's highest qualification
- Matched Skills — Skills from the candidate's resume that match the job requirements
- Missing Skills — Required skills not found in the candidate's resume
- Status — Current application status (Pending, Shortlisted, Reviewed, Rejected)

💡 How scoring works: The AI calculates: Overall Score = (Education Score $\times$ 40%) + (Skill Score $\times$ 60%). Skill Score = matched skills $\div$ total required skills. Education Score = 1 if the candidate meets or exceeds the required education level, 0 otherwise.

4.4 Exporting Candidates to CSV

Step 1 On the Candidate Ranking page for a job, click the Export CSV button.

Step 2 A CSV file will be downloaded automatically to your browser's default download folder.

The CSV contains the following columns: Rank, Candidate Name, Email, Job Title, Overall Match Score (%), Skills Score (%), Education Score (%), Status, and Applied Date.

 **Tip:** The CSV export is ideal for sharing shortlisted candidates with hiring managers or storing records outside the system.

5. Candidate Portal

The Candidate Portal is accessible only to users registered with the Candidate role. It allows candidates to build their profile, upload their resume, apply for jobs, and track their application outcomes.

5.1 Managing Your Profile

Your profile contains the information the AI uses to supplement resume analysis. Keeping your profile up to date improves the accuracy of your match scores.

Step 1 Click Profile in the sidebar navigation.

Step 2 Fill in or update the following fields:

Full Name	Your full name as it should appear to HR
Phone	Your contact phone number
Education Level	Your highest completed qualification: Diploma, Bachelor's, Master's, or PhD
Skills	Add each skill and press Enter or comma to tag it (e.g. Python, React, SQL)
Experience	Years of relevant work experience

Step 3 Click Save Profile to apply changes.

The screenshot displays the 'My Profile' page in the AI Resume Matcher application. The page has a dark theme and a sidebar on the left with navigation links: 'Browse Jobs', 'My Resume', 'My Applications', 'My Profile' (highlighted), and 'Logout'. The main content area is titled 'My Profile' and includes a subtitle 'Manage your personal information, career details and skills.' Below this, there are several sections: a profile card for 'John Doe' with a 'Candidate' role, a 'Resume' section showing an uploaded resume, and a 'QUICK STATS' section with 'Applications' (3). The 'ACCOUNT INFO' section (read-only) includes fields for 'Full Name' (John Doe), 'Email' (john.doe@test.com), and 'Phone' (+60123456789). The 'CAREER INFORMATION' section (editable) includes a dropdown for 'Education Level' (Bachelor's Degree), a field for 'Years of Experience' (4), and a 'LinkedIn URL' (https://linkedin.com/in/yourname). There is also a 'Skills' section with a list of skills and a 'Total years of work experience' label.

Figure: Candidate Profile page

5.2 Uploading Your Resume

Your resume is the primary input for the AI scoring engine. Always upload the most up-to-date version before applying for a job.

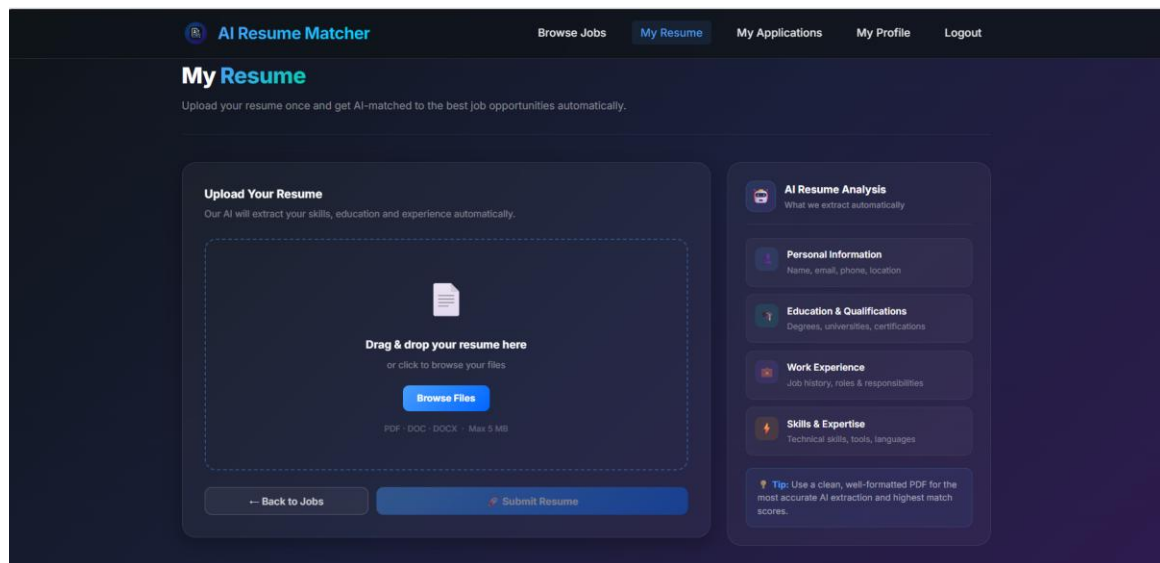
Step 1 Navigate to your Profile page.

Step 2 In the Resume Upload section, click Choose File.

Step 3 Select your resume file. Accepted formats: PDF, DOCX, or TXT. Maximum recommended file size: 5 MB.

Step 4 Click Upload Resume.

A confirmation message will appear once your resume has been uploaded successfully. Your previously uploaded resume will be replaced.



Tip: PDF format is recommended as it preserves formatting best. Ensure your resume clearly lists your skills and education level — the AI extracts these from the document text.

5.3 Browsing Job Listings

Step 1 Click Jobs in the sidebar to open the job listings page.

Step 2 All currently active job postings are displayed, each showing the job title, description excerpt, and required skills.

Step 3 Use the Search bar to filter jobs by title or keyword.

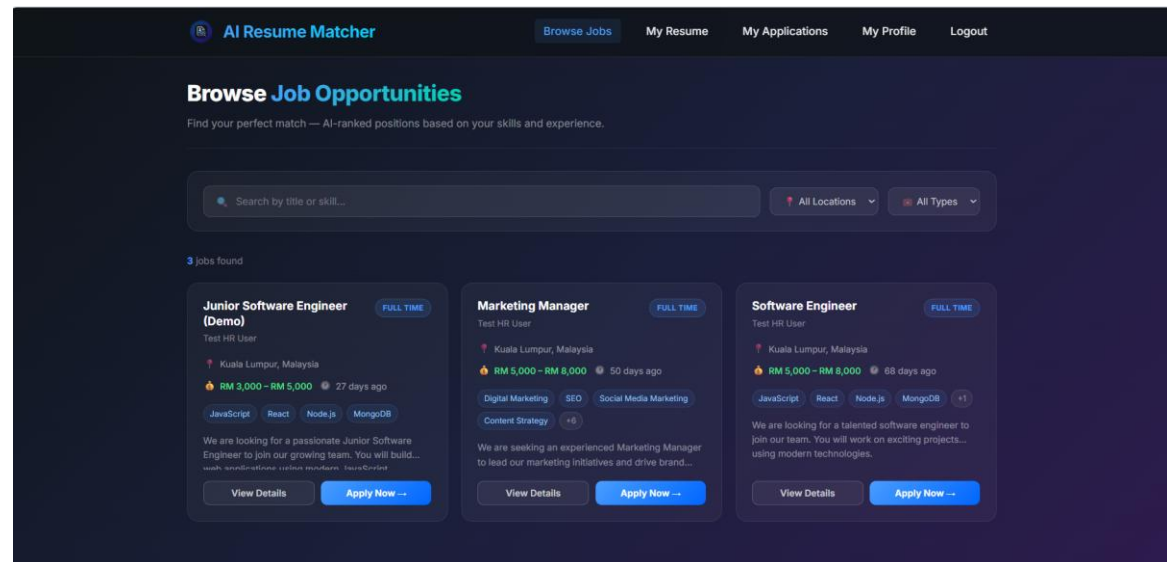


Figure: Candidate Job Listings page — active jobs with search bar

5.4 Applying for a Job

Step 1 Click on a job title to open the job detail view.

Step 2 Review the full job description, required skills, and minimum education level.

Step 3 Click Apply Now.

The system will immediately submit your application and trigger the AI module to analyse your uploaded resume against the job requirements. Your match score is computed and stored at the point of application.

⚠ Important: You must have uploaded a resume before applying. If no resume is on file, the application will still be submitted but without an AI match score.

5.5 Tracking Your Applications

- Step 1** Click My Applications in the sidebar.
- Step 2** All your submitted applications are listed, showing the job title, company, applied date, your AI match score, and status.
- Step 3** Use the Status Filter at the top to view applications by: All, Pending, Shortlisted, or Rejected.

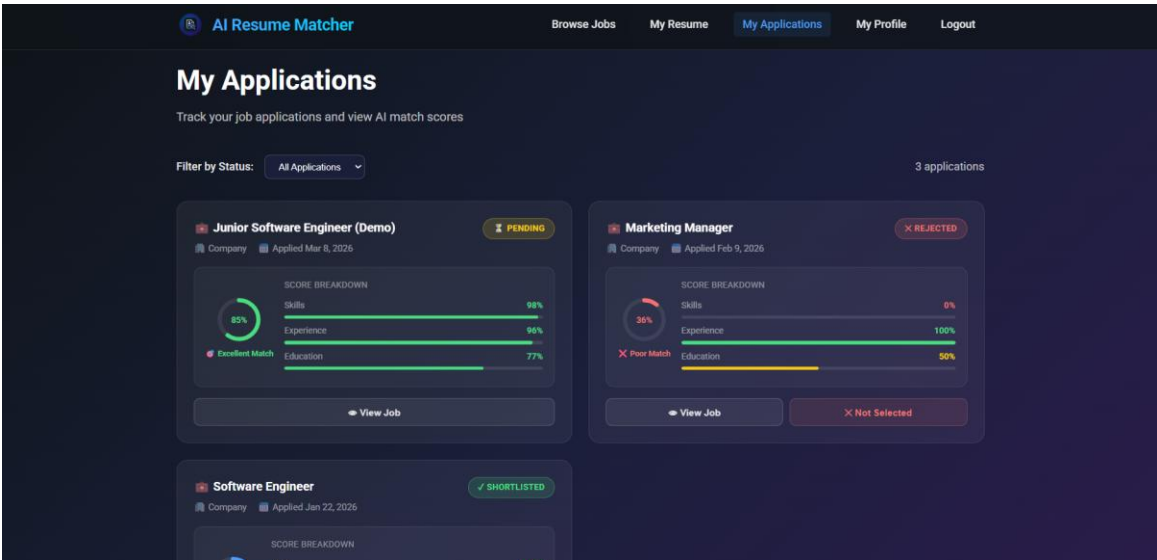


Figure: My Applications page — application list with match scores and status filter

Application status meanings:

Pending	Your application has been received and is awaiting HR review
Shortlisted	HR has shortlisted you for this role — congratulations!
Rejected	HR has decided not to progress your application for this role at this time

6. Understanding the AI Match Score

6.1 How the Score is Calculated

When a candidate applies for a job, the AI module performs the following steps:

- Resume Parsing — The system extracts raw text from the uploaded resume file (PDF, DOCX, or TXT).
- Skill Extraction — spaCy's PhraseMatcher identifies skills in the resume text by comparing against a pre-defined skill library.
- Education Classification — Rule-based keyword detection identifies the candidate's highest education level (Diploma, Bachelor's, Master's, or PhD).
- Score Calculation — A weighted formula is applied:

```
Overall Score = (Education Score × 0.40) + (Skill Score × 0.60)
Skill Score    = Matched Skills ÷ Total Required Skills
Education
Score = 1.0 if candidate meets required education level, 0.0 otherwise
```

6.2 Score Colour Coding

Match scores are colour coded on the HR Candidate Ranking page for at-a-glance interpretation:

● Green (≥70%)	Strong match — candidate meets most or all requirements
● Amber (40–69%)	Partial match — candidate meets some requirements; review recommended
● Red (<40%)	Weak match — candidate does not meet minimum requirements

6.3 Improving Your Match Score (Candidates)

- Ensure your resume explicitly mentions the skills listed in the job posting.
- Keep your Profile skills list updated — this provides an additional signal to the AI.
- Use industry-standard skill names (e.g. "JavaScript" not "JS coding") for better pattern matching.
- Ensure your education level is clearly stated in your resume.

7. Troubleshooting

7.1 Common Issues

Issue	Likely Cause	Resolution
Cannot connect to backend	Backend server is not running	Run <code>node server.js</code> in the <code>/backend</code> directory. Check for errors in the terminal.
AI module returns an error on application	AI module is not running	Run <code>uvicorn main:app</code> in the <code>/ai-module</code> directory. Ensure the virtual environment is activated.
Match score shows as 0% or N/A	No resume uploaded, or resume could not be parsed	Upload a resume from your Profile page. Ensure the file is PDF, DOCX, or TXT and is not password-protected.
Dashboard chart not loading	Browser cached old data, or backend is slow	Hard-refresh the page (Ctrl + Shift + R). Ensure MongoDB is running.
Login fails with correct credentials	MongoDB is not running	Start MongoDB. Check the <code>MONGO_URI</code> in your <code>.env</code> file matches your local MongoDB connection string.
Candidate status not updating on candidate portal	Browser serving stale cached data	Hard-refresh the candidate portal page (Ctrl + Shift + R).
CSV export is empty	No candidates have been shortlisted for this job	Export includes all candidates. Check you are on the correct job's candidate page.

7.2 Checking Service Health

You can verify each service is running correctly by visiting:

- Backend health check — <http://localhost:5000/health> (should return JSON with status: ok)
- AI module health check — <http://localhost:8000/health> (should return JSON with status: ok)
- AI interactive docs — <http://localhost:8000/docs> (FastAPI Swagger UI)

8. Glossary

AI Module	The Python/FastAPI microservice responsible for parsing resumes and computing match scores
ATS	Applicant Tracking System — software used by organisations to manage recruitment workflows
bcrypt	A password hashing algorithm used to securely store user passwords in the database
CSV	Comma-Separated Values — a plain text file format for tabular data, openable in Excel
Education Score	A component of the match score: 1.0 if the candidate meets the required education level, 0 otherwise
FastAPI	A modern, high-performance Python web framework used to build the AI microservice
HR Portal	The web interface used by HR professionals to manage jobs and review candidates
JWT	JSON Web Token — a secure token issued on login, used to authenticate API requests
Match Score	An AI-computed percentage (0–100%) indicating how well a candidate matches a job posting
MongoDB	The NoSQL document database used to store users, jobs, and application data
NLP	Natural Language Processing — the branch of AI used to extract meaning from text (e.g. resumes)
Skill Score	A component of the match score: $\text{matched skills} \div \text{total required skills}$
spaCy	An NLP library used for skill extraction via pattern matching on resume text
TF-IDF	Term Frequency-Inverse Document Frequency — a text relevance scoring technique used in the AI module

9. Document History

Version	Date	Author	Changes
1.0	31/03/2026	[Your Name]	Initial release for FYP submission